

Alluri Jeswanth Joel

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Education

Indian Institute of Information Technology (IIIT) Kottayam, Kerala, India

Nov 2022 – May 2026

B.Tech – Computer Science and Engineering

Coursework: Operating Systems, Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Databases, Cloud Computing

Projects

Neural OPS – Enterprise AI Hiring & Candidate Intelligence Platform

GitHub

FastAPI, Python, PostgreSQL, Redis, Qdrant, Gemini, Celery, MLflow, Docker, Next.js · May 2026

- Designed a hybrid multi-tenant ranking architecture combining Qdrant vector similarity, skill-match scoring, and experience signals; orchestrated async ingestion and embedding via Celery workers with experiments tracked in MLflow.
- Benchmarked candidate-matching workloads across **241 telemetry runs** under concurrent stress testing, achieving a median matching latency of **28.31 ms**.
- Implemented Redis-backed concurrency controls and LLM budget governance; instrumented telemetry pipelines for latency benchmarking under concurrent workloads.

Simplify – Citation-Enabled RAG Platform

GitHub / Live

Next.js, TypeScript, FastAPI, Python, MongoDB Atlas, Pinecone, Supabase, Gemini · April 2026

- Built an async document ingestion pipeline covering file validation, text extraction, overlap chunking, Gemini embedding generation, and Pinecone vector upserts; benchmarked **542 ingestion runs** with median latency **24.20 ms** and p95 below **353 ms**.
- Integrated citation-backed response generation with Gemini streaming; secured with JWT auth (refresh-token rotation, access-token denylisting) and email OTP with rate limiting.
- Deployed frontend on Vercel and FastAPI backend on Render; persisted metadata in MongoDB Atlas with Supabase Storage for raw file management.

TaskFlow – High-Performance Task Management API

GitHub

Java 17, Spring Boot 3, Spring Security, JWT, PostgreSQL, Docker, GitHub Actions · 2025

- Achieved **p95 request latency below 50 ms** and built JWT-secured REST APIs with RBAC (ADMIN/USER) and JPA Specification-based dynamic filtering across status, priority, assignee, and keyword fields.
- Implemented an immutable TaskHistory audit trail logging every field-level mutation with actor and timestamp; instrumented benchmark telemetry across auth and request pipelines.
- Containerized with multi-stage Docker builds; GitHub Actions CI/CD runs JUnit 5 + Mockito tests and publishes Swagger UI on every push.

Technical Skills

Languages	Java, Python, C/C++, JavaScript, TypeScript, SQL (PostgreSQL, MySQL), R
Web	Next.js, React, Node.js, Express.js, FastAPI, Flask, Spring Boot
AI / ML	LangChain, Pandas, NumPy, Scikit-learn, RAG, Vector Search
MLOps & Tools	MLflow, Celery, Docker, GitHub Actions
Databases	PostgreSQL, MongoDB Atlas, Redis, Pinecone, Qdrant, Supabase, FAISS, ChromaDB
Cloud & Deployment	GCP, Gemini (Chat + Embeddings), Vercel, Render

Certifications

Google Cloud Study Jam – GDSC, IIIT Kottayam

Oct 2023

Completed hands-on labs: Compute Engine, IAM, Cloud Storage, and Kubernetes Engine workloads.

Graph Theory Programming Camp – AlgoUniversity

2024

Solved 17+ graph problems spanning BFS/DFS variants, shortest-path algorithms, and DP on graphs.

Achievements & Leadership

Organiser – Adavya Freshers Fest 2024 | Outreach – Sponsorship

2024

Coordinated sponsor outreach and contributed to external partnership efforts for the college-wide freshers event.